

#b nodes with a given in-degree
 $n(K_b; t+1) = n(K_b; t)$

+ $m \cdot (1-g) \cdot \Pi_b(K_b-1, t) \cdot n(K_b-1, t) \iff$ (# promotions opportunities available $\cdot$ prob of promotion assigned for this level of skill & type of person $\cdot$ # ppl)

- $m \cdot (1-g) \cdot \Pi_b(K_b, t) \cdot n(K_b, t)$ (1)

+ $(1-g) \delta_{K_b, 0}$ (in BA new node has $K_a^{(in)} - K_b^{(in)} = 0$) (normal, suit, weighted expected value ansatz)

prob("b" node receives new in-degree from new node) = $\Pi_{b,i} = \frac{(1-g)K_{a,i} + g(1-g)K_{b,i}}{Z_b(t)} = \frac{\lambda_b K_{a,i} + \mu_b}{Z_b(t)}$

$\sum_{i \in V} (\Pi_{b,i} + \Pi_{a,i}) = 1$ as new node certainly gives some in-degree when $m > 0$

$$\Rightarrow \sum_{i \in V_b} \Pi_{b,i} + \sum_{i \in V_a} \Pi_{a,i} = 1 \quad (*)$$

($\iff$ prob("b" node gets in-degree) + prob("a" node gets in-degree) = 1)

$$\iff \sum_{i \in V_b} K_{b,i} = \lambda_b E_b(t)$$

$$\mu_b \sum_{i \in V_b} 1 = \mu_b N_b(t)$$

for simplicity let: $N(t_{init}) = t_{init}$ & $E^{tot, in}(t_{init}) \cdot m t_{init} = E_a^{in} + E_b^{in}$

lets assume $E_b^{in}(t_{init}) = m(1-g)ht + g(1-g)ht_{init} = m t_{init}$

$$\lambda_b(t) + \mu_b(t) = N(t) \quad N_b(t) = (1-g)N(t)$$

& that since linear?

$$\Rightarrow E_b(t) = \lambda_b m t$$

$$\& N_b(t) = (1-g)t$$

symmetric arg for "a":

$$\Rightarrow E_a(t) = m[gh + (1-g)(1-h)]t = m \lambda_a t$$

$$\Rightarrow (*) \rightarrow \frac{(m \lambda_b + (1-g) \mu_b)t}{Z_b(t)} + \frac{(m \lambda_a + g \mu_a)t}{Z_a(t)} = 1$$

The point of Z_a, Z_b is to ensure (*)

if we choose $Z_a = Z_b$ then the result is still

true (?)

$$\Rightarrow Z(t) = [m(\lambda_a^2 + \lambda_b^2) + (g \mu_a + (1-g) \mu_b)] t$$

$$| \text{it's useful here to motivate } m(1-g)[m(\lambda_a^2 + \lambda_b^2) + (g \mu_a + (1-g) \mu_b)]' = \tilde{Z}$$

$$\Rightarrow (1) \rightarrow \therefore n(K_b, t+1) = n(K_b, t)$$

$$+ \tilde{Z} t' (\lambda_b(K_b-1) + \mu_b) n(K_b-1, t)$$

$$- \tilde{Z} t' (\lambda_b K_b + \mu_b) n(K_b, t) \quad (2)$$

$$+ (1-g) \delta_{K_b, 0}$$

the part
 these are E
 doesn't really work
 in the code.

$$\left\{ \begin{aligned} &(\lambda_a + \lambda_b = hg + (1-h)(1-g) + h(1-g) + (1-h)g \\ &= (2hg + 1 - h - g) + (h + g - 2hg) \\ &= 1) \text{ nice sanity check} \\ &\Rightarrow E^{tot, in} = E_a + E_b \\ &\lambda_a^2 + \lambda_b^2 = \text{horrible} \dots \end{aligned} \right.$$